Aariz Khan

B.Tech - Artificial Intelligence & Machine Learning

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AI and machine learning undergraduate building retrieval and NLP systems: document ingestion, embeddings and vector search in PostgreSQL, hybrid retrieval with reranking, and supervised models evaluated on public datasets.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning

CMR University, Bengaluru | 2023 - 2027 | CGPA 8.8

PROJECTS

AARIZ AI - Retrieval-Augmented Document Chatbot

In progress | Next.js, React, TypeScript, Supabase, PostgreSQL + pgvector, Ollama, Gemini | github.com/Aariz-a-z/aariz-ai

- Built the full retrieval path: text extraction from PDF and HTML, semantic chunking at ~600-900 tokens with ~100 token overlap, embedding, and storage in PostgreSQL with pgvector.
- Implemented hybrid retrieval combining vector similarity with full-text search, fused using Reciprocal Rank Fusion, then reranked; reranking is fail-open, returning the fused ordering rather than failing the request.
- Built a provider abstraction selecting local (Ollama) or hosted (Gemini) models by environment variable, for both generation and embeddings.

AI Resume Analyzer

Python, Streamlit, Sentence-Transformers, PyMuPDF, Plotly | github.com/Aariz-a-z/AI-RESUME-ANALYZER

- Scores a resume against a job description by encoding both with all-MiniLM-L6-v2 and comparing them by cosine similarity.
- Extracts and normalises PDF text with PyMuPDF, flagging documents an ATS is likely to mis-parse, and reports missing skills from an alias ontology as a radar chart.

Sales Forecasting

Python, pandas, NumPy, scikit-learn | github.com/Aariz-a-z/Sales-forecasting

- Aggregated daily transactions into a monthly series with trend, cyclical month and lag features for linear regression; evaluated on a chronological hold-out rather than a random split. R-squared: 0.525.

Heart Stroke Prediction

Python, pandas, scikit-learn | github.com/Aariz-a-z/Heart-stroke-prediction

- Logistic regression on an imbalanced public dataset (~5% positive) with balanced class weights and leak-free scaling; evaluated by confusion matrix, per-class recall and ROC-AUC of 0.839 on a stratified 80/20 hold-out.

Wine Quality Classification

Python, scikit-learn | github.com/Aariz-a-z/Wine-quality-classification

- Multiclass SVM over standardised physicochemical features, evaluated per class with a confusion matrix rather than on aggregate accuracy.

Iris Flower Classification

Python, scikit-learn | github.com/Aariz-a-z/Iris-flower-classification

- K-nearest neighbours with feature scaling and a stratified split, plus a sweep over k from 1 to 20.

Travel Planner

HTML, CSS, PHP, MySQL | github.com/Aariz-a-z/project-1

- Eight-page travel planning interface with PHP sign-up and sign-in validating against a MySQL database, and a CSS-only navigation menu.

Python Fundamentals

Python | github.com/Aariz-a-z/python-projects

- Password generator, student management system, authentication with hashed passwords, and an expense tracker; all file-backed.

SKILLS

Programming & Tools: Python, TypeScript, SQL (Basics), Git, GitHub, Jupyter, Google Colab, VS Code

AI & NLP: Retrieval-augmented generation, embeddings, semantic search, Sentence-Transformers, vector search (pgvector)

Machine Learning & Data: scikit-learn, regression (linear, logistic), classification (KNN, SVM), model evaluation, feature engineering, pandas, NumPy, Matplotlib, seaborn, EDA

Web & Deployment: Next.js, React, Streamlit, HTML, CSS, Supabase, PostgreSQL